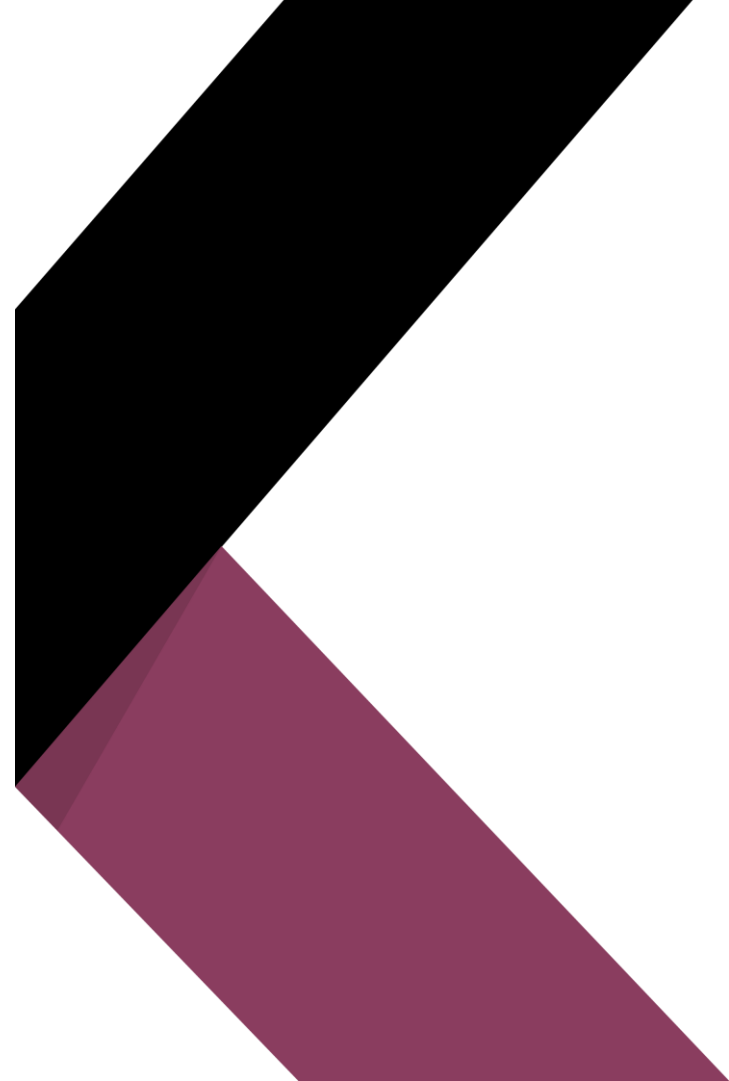


Functional

Specifications





Functional Specifications

What is a functional specification

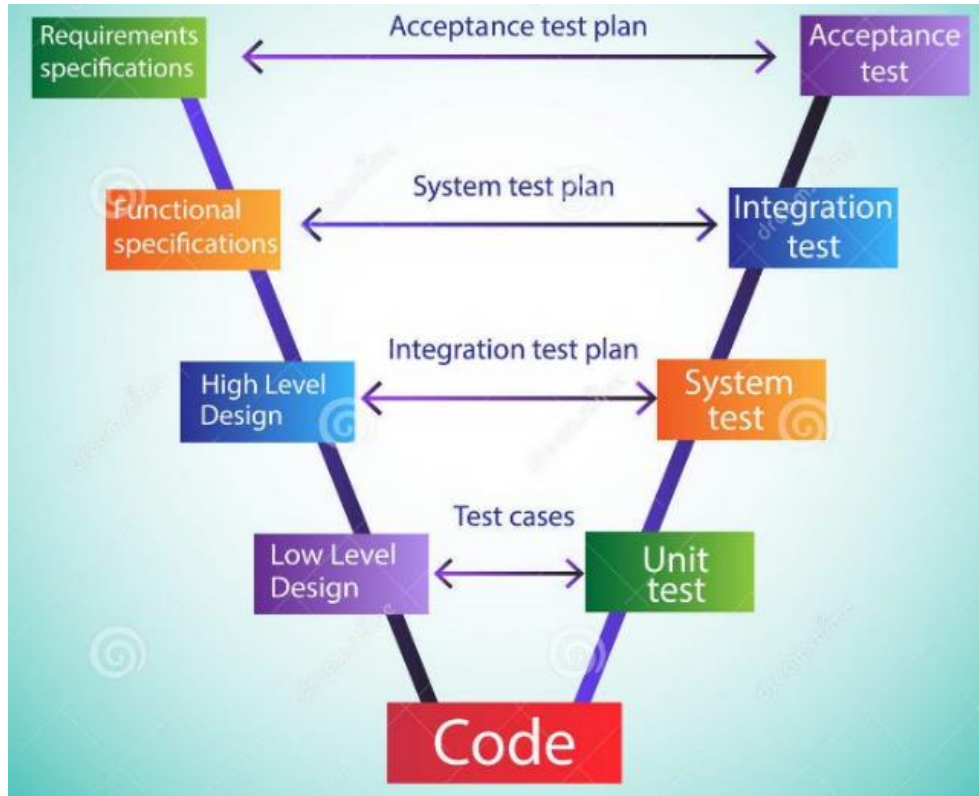
Functional

Specification

- A functional specification, or VF (Vehicle Function), is a document written by a professional figure, called a functionalist, in order to have a complete view of the behavior of a given function.
- In a VF lists any changes made with respect to any previous versions (if there are any), the available features, the structure, the various nodes of the vehicle that are involved, the expected behavior in different environmental conditions, diagnosis and eventual recovery adopted .
- A VF is not related to the component structure.
- Starting from a VF, the software is written.



Positioning in the V-Model





Possible functionalities in a car

For each of the following features, multiple VFs can be written

- Car access
- Theft Protection
- Guide Aid
- Climate
- Vehicle dynamics
- Management of the electrical and electronic system
- Braking
- Energy generation and management
- Motopropulsion management
- Interior lighting
- Lighting and external signaling
- Servo assisted movement
- Passive security
- Visibility and wiping
- infotainment
- Instrument panel functions
- NIT



Apparance

and structured as a VF - Nomenclature

A VF is presented as a document to which a name is given, referred to the function object of the document, followed by an alphanumeric series.

V stands for Version and **R** for Release; the Version allows you to version the functionality according to the set-up, while the Release allows you to take into account the changes made to the VF in subsequent releases (adaptations to new projects for example). The name is usually composed of:

Function Name[Fx.**V**fx **R**x]

Below an example of the name of a VF related to the manettino

Manettino Management [F165.Vf14_R4]



Apparance

and structured as a VF - Revision Notes

The first part of a VF consists of the 'Revision Notes' which is inserted, by date, the owner of the VF, if it is derived from a project different from that for which it is working (porting) and in this case it is reported the name of the project from which the VF originates and if there are any, the changes made are listed. An example is shown on the side.

1.1 Revision notes

24/02/2014 (Molinari): New release for F151M derived from F142M L7

12/03/2014 (Molinari):

- Functional diagram modified:
 - NQS reads directly on C-CAN the NCSFailSts and SuspensionmodeSts signals
 - NQS reads directly on C-CAN the TC/ASRControlSts signal from F134.NFR
 - NQS reads directly on C-CAN the NCDFailSts and DIFF_BuzzReqSts signals
 - NCR reads EngineSts and StopStartSts signals on PWT-CAN Network
 - F175.NQS reads ManettinoSts and ManettinoFailSts from F165.NBC on C-CAN in according to NFR,NCM,NCR,NCS systems
- Function revision update from R2 to R3

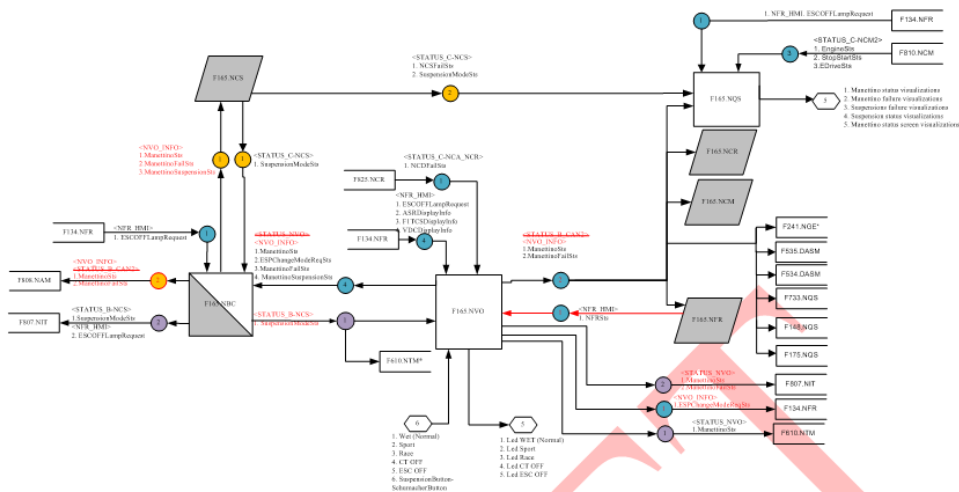
04/12/2017 (Mosca): F173 RMF015 New NVO introduction (Manettino)

- Functional diagram modified
 - Added NVO_INFO message on C1-CAN network
 - Replaced STATUS_B_CAN2 BH CAN message with NVO_INFO BH CAN message
 - Added NFR_HMI.NFRsts C1 CAN signal from F165.NFR to F165.NVO
 - Added ManettinoSts and ManettinoFailSts BH CAN signals from F165.NVO to F807.NIT
 - Added ESPChangeModeReqSts C1 CAN signal from F165.NVO to F134.NFR
- Function revision modified from Vf14_R3 to Vf14_R4



Appearance

The functional diagram graphically describes all the interactions related to the function in question. All connections between the involved nodes and the related CAN / LIN messages that are exchanged are reported.



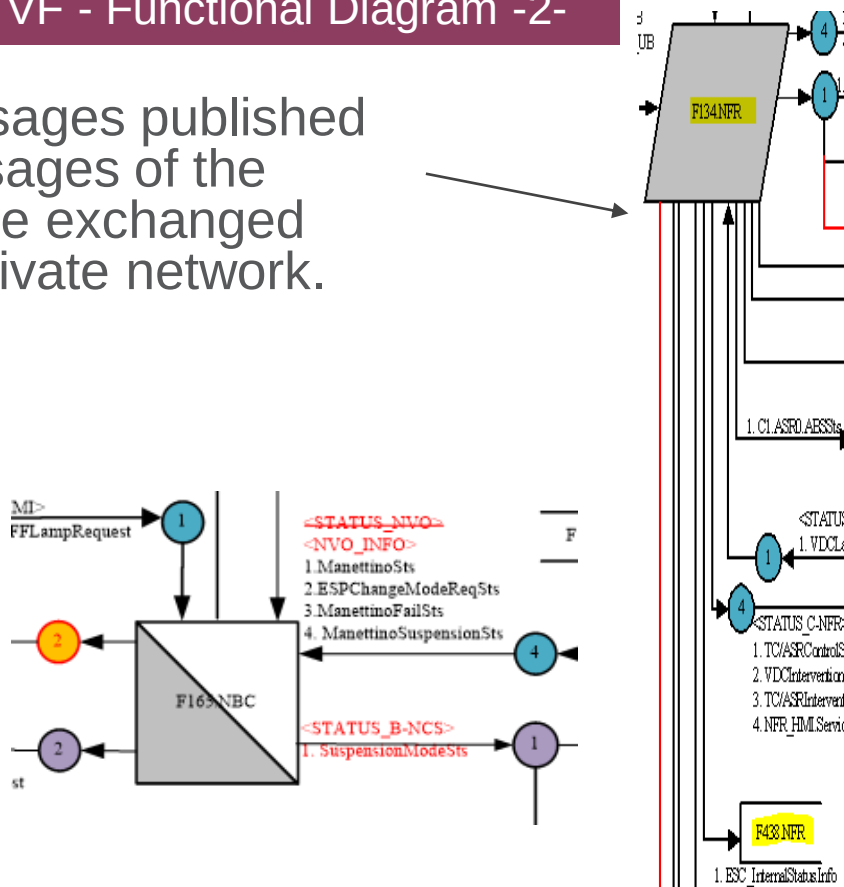
(*) only if NTM is present

Appearance

and structured as a VF - Functional Diagram -2-

In addition to the CAN / LIN messages published on the network, the internal messages of the component are reported which are exchanged with the internal modules via a private network.

Usually different colors are used to identify different networks such as C1CAN, C2CAN, C3CAN, BHCAN, PCAN, LIN and on each connection branch the name of the message and the transmitted and / or received signal are reported.





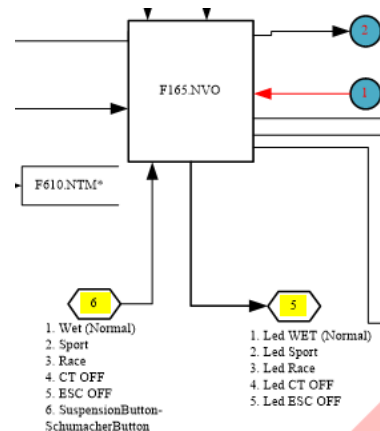
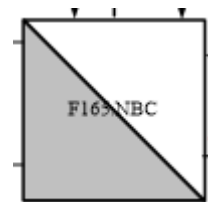
Apparance

and structured as a VF - Functional Diagram -3-

The nodes drawn with two different colors are those nodes that also act as gateways, which can in turn be direct or indirect.

The numbers present in the blocks that identify the nodes refer to the VF from which or to which the signals are transmitted and / or received.

The outputs of the nodes are not necessarily directed towards other nodes, but could be acoustic and / or visual signals directly perceived in the vehicle by the user, for example an LED.





Apparance

and structured as a VF - Working conditions

The working conditions describe the conditions under which the function is operational. You can have more conditions such as key-on and key-off, or only key-on, etc., this depends on the function.

The operating status for the manettino is shown below.

1.3 Working conditions

Working mode	Yes	No	Remarks
Key off (+30)		X	
Key on (+15)	X		
<u>Timed</u> , from key off (*)		X	
<u>Excluded during crank</u> (+)		X	
<u>Excluded with battery</u> <u>discharged</u> (+)		X	

(*)Write the timing expressed in minutes in the field "Remarks"

(+)Write the mode (SW or HW) of function exclusion in the field "Remarks".



Apparance

and structured as a VF - Function description -1-

The section dedicated to 'Function description' is the heart of VF because it describes in detail the actual operation of the function. The first step lists the nodes involved in the specific function that 'make logic' and not, and for each of them the relative signals are indicated in input and / or output.

In the description, if present, state machines are also inserted relating to particular operating conditions.

An example is shown below.



Apparance

and structured as a VF - Function description -2-

Suspension button Led Management

The suspension button is provided of a status LED. The LED is turned on when SuspensionModeSts = Comfort

Screen "Status Manettino" Management

The screen "Status Manettino" is recallable through the pressure of the VDA button in every manettino position. This screen contains information on the systems that depend from manettino's state and it is activated in two modalities: fixed and timed.

In case of failure on Manettino, this screen isn't recallable and visualized.

For further details to see the specification finalized NQS.

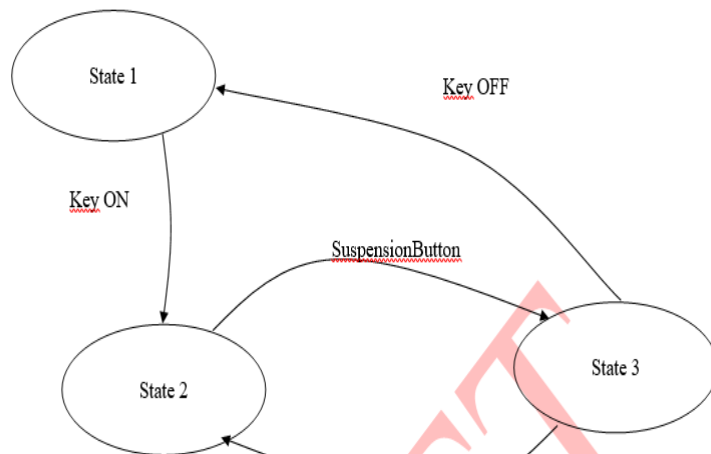
Safety in case of failures

If the message of NVO is missing on CAN, NBC will send to the various systems the WET mode and will set the bit ManettinoFailSts. The WET mode will remain until Key Off; at the following Key ON, if the failure still exists, NBC will send to the network the WET mode and the bit ManettinoFailSts=1.

When NFR receives the bit ManettinoFailSts=1, it will activate the ESC if it was previously disengaged.

If the message of NBC is missing on CAN, the systems will be calibrated on WET (Normal).

The steering wheel node must be in WET conditions and lead the engagement of the ESC, if disengaged, when it detects a mistake on the selector.



State1: Key OFF (Suspension's state is slaved of manettino)

State2: Key ON (Suspension's state is slaved of manettino)

State3: Key ON (Suspension's state is slaved of suspension button)



Apparance

and structured as a VF - Working Characteristics

The 'Working characteristic' lists the function 'characteristics' of the function understood as the proxy values to be used for that specific function to be used.

There are also information related to the activation of the operation.

1.6 Working characteristics

<u>Characteristics</u>	<u>Yes</u>	<u>No</u>	<u>Description</u>
<u>Required EOL (1)</u>	x		Start Byte:29;Start Bit:3;Used by:NOS (NCS absent / present);Checked by;; Start Byte:61;Start Bit:7;Used by:NOS (S&S absent / present);Checked by;; Start Byte:58;Start Bit:1;Used by:NOS (Coupe / Spider);Checked by;;
<u>By event management(2)</u>	x		<u>Selection Manettino</u> <u>Suspension button pressure</u>
<u>Periodical management</u>		x	

(1)Write the parameters, nodes and default values in the field "Description".

(2)Write the node and the cause leading to the control by event of the function in the field "Description".





Apparance

and structured as a VF - CAN WakeUP

The section dedicated to the CAN WakeUP describes if this function, once activated, can wake up the CAN network, as shown in the example table.

1.7 Wake-up generation for CAN communication



		Event and node generating the wake-up			
Yes	No	Ext	Int.	Node	Description
	x				





Apparance

and structured as a VF - Diagnosi e Recovery

The last part of a VF is dedicated to the diagnosis and to the recovery, if present.

Usually if the diagnosis is present for a function, this is described in this section. For further information, we recommend viewing the CDD, a document dedicated only to the diagnosis in which all the relative DTCs and the validation / invalidation conditions are meticulously described.

1.8 Diagnosis of the function

1.8.1 Discrete OUTPUT

Output	<u>Diagnosis availbale</u>	Active Diagnosis	<u>Node/ Comp.</u>
<i>Buzzer (if present)</i>	NO	Yes	NQS
<i>Suspension failure</i>	NO	Yes	NQS
<i>Signals Manettino</i>	NO	Yes	NQS

For details refer to SDF of the component.



A key tool that goes hand in hand with the writing of a VF is the DBC file that is mapped for each CAN protocol, the entire network of posts and the related signals. In the next slide an example of a DBC file related to the BHCAN network.



Appendix

DBC File -2-

Vector CANdb++ Editor - C:\Users\user\Downloads\DB_SR_BHCAN_F173_2018_01_30_Rev173 (1).dbc - [Overall View]

File Edit View Options Window Help

Networks

- ECUs
 - AMP
 - BCM
 - Environmental variables
 - BCM
 - DIAG_TESTER
 - DVD
 - LBSS
 - NAG
 - NAP
 - NCL
 - NCP
 - NDC
 - NDR
 - NIT
 - NPS
 - NQS
 - NRC
 - NSP
 - NSW
 - NTM
 - NTP
 - NVO
 - PIP
 - RBSS
- Environment variables
- Network nodes
- Messages
- Signals
 - _3WaysValveSts
 - ABSDisplayInfo
 - ABSLamp_FailSts
 - ACC_DrivingMode

Name	Message	Multiplexing/G...	Startbit	Leng...	Byte Order	Value Type	Initial Value	Factor	Offset	Mini...	Max...
ABSDisplayInfo	NFR_HMI	-	4	1	Motorola	Unsigned	0	1	0	0	1
ACC_DrivingMode_fbK	STATUS_DASM	-	13	1	Motorola	Unsigned	0	1	0	0	1
ACC_RadarBlinded	STATUS_DASM	-	58	1	Motorola	Unsigned	0	1	0	0	1
ACC_SetSpeedKPH	STATUS_DASM	-	24	8	Motorola	Unsigned	0	1	0	0	2
ACC_SetSpeedMPH	STATUS_DASM	-	32	8	Motorola	Unsigned	0	1	0	0	2
ACCFailSts	STATUS_DASM	-	47	1	Motorola	Unsigned	0	1	0	0	1
ACCSystemSts	STATUS_DASM	-	60	3	Motorola	Unsigned	0	1	0	0	7
ActiveLoadMaster	NWM_NBC	-	10	1	Motorola	Unsigned	0	1	0	0	1
ActualGearForDisplay	STATUS_B_NCA_NCR	-	17	4	Motorola	Unsigned	0	1	0	0	1
ActualSpeed	STATUS_AUWP_HVB	-	8	8	Motorola	Unsigned	7140	28	0	0	7
AHB_ON	HAPTIC1	-	43	1	Motorola	Unsigned	0	1	0	0	1
AHBFault	HAPTIC1	-	42	1	Motorola	Unsigned	0	1	0	0	1
AHBFeature	EXTERNAL_LIGHTS	-	10	1	Motorola	Unsigned	0	1	0	0	1
AlarmSysTurnOn	STATUS_NBC	-	34	5	Motorola	Unsigned	0	1	0	0	3
ASBM_LedControlSts	HAPTIC1	-	30	2	Motorola	Unsigned	0	1	0	0	3
ASRDisplayInfo	NFR_HMI	-	14	2	Motorola	Unsigned	0	1	0	0	3
AutoFeedbackLed	EXTERNAL_LIGHTS	-	6	2	Motorola	Unsigned	0	1	0	0	3
Automatic_RoboticXmsnFailSts	STATUS_B_NCA_NCR	-	16	1	Motorola	Unsigned	0	1	0	0	1
AutomaticRoboticTransmissionDegr	STATUS_B_NCA_NCR	-	40	1	Motorola	Unsigned	0	1	0	0	1
AutomaticUpReqCntrl	STATUS_NPG	-	41	1	Motorola	Unsigned	0	1	0	0	1
AutoParkEnable	STATUS_B_NFR	-	14	1	Motorola	Unsigned	1	1	0	0	1
AVHActive	STATUS_B_NFR	-	15	1	Motorola	Unsigned	0	1	0	0	1
AVHDisplayInfo	NFR_HMI	-	0	2	Motorola	Unsigned	0	1	0	0	3
AXSStat	Gyro_Data	-	10	2	Motorola	Unsigned	3	1	0	0	3
AYStat	Gyro_Data	-	26	2	Motorola	Unsigned	3	1	0	0	3
BatteryChargerPresent	EXTERNAL_LIGHTS	-	5	1	Motorola	Unsigned	0	1	0	0	1
BatteryVoltageLevel	STATUS_B_NCM2	-	24	7	Motorola	Unsigned	0	0.16	0	0	1
BatteryVoltageLevel	STATUS_NBC	-	24	7	Motorola	Unsigned	0	0.16	0	0	1
BatteryVoltageLevelFailSts	STATUS_B_NCM2	-	39	1	Motorola	Unsigned	0	1	0	0	1
BonnetSts	STATUS_NBC	-	15	1	Motorola	Unsigned	0	1	0	0	1

ECU - Node: BCM [BCM]

Ready

NUM



Another fundamental tool that goes hand in hand with writing a VF is the LIN file in which the entire message network and the related signals are mapped for the LIN protocol. In the next slide an example of a LIN file related to the wiping functionality.



Appendix

LIN File -2-

File Edit Help

Frames/Commands

Nodes/Frames

Ferrari_NBCAlarm-CSP_SignalList_LIN1_F15

Unconditional Frames

- Command_ASU (0)
- Heart_Beat_ASU (2)
- Ins_Disins (1)
- Status_BCM (3)
- Heart_Beat_Feedback_ASU (17)
- Status_ASU (18)
- FSdata (35)
- RLSData (41)**
- SoSe (50)
- ComToRLS (38)
- RLSCfgData (29)

Diagnostic Frames

- MasterReq (60)
- SlaveResp (61)

Event-Triggered Frames

Sporadic Frames

Predefined Commands

- AssignNAD
- ConditionalChangeNAD
- AssignFrameId
- UnassignFrameId
- ReadByIdentifier
- DataDump
- FreeFormat

Schedule Tables

Table / Slot	Type	Delay [ms]	Max. Frame Time [ms]	Min. Frame Time [ms]	Act. Min/Max Period [ms]
Operating_Armed					
Status_BCM (3)	Unconditional	10.00	6.12	4.37	100.00 / 100.00
MasterReq (60)	Diagnostic	20.00	9.05	6.46	100.00 / 100.00
Heart_Beat_ASU (2)	Unconditional	10.00	5.39	3.85	100.00 / 100.00
Status_ASU (18)	Unconditional	30.00	4.66	3.33	100.00 / 100.00
Heart_Beat_Feedback_ASU (17)	Unconditional	10.00	3.94	2.81	100.00 / 100.00
SlaveResp (61)	Diagnostic	20.00	9.05	6.46	100.00 / 100.00
Operating_Armed_InsDisins					
Status_BCM (3)	Unconditional	10.00	6.12	4.37	100.00 / 100.00
MasterReq (60)	Diagnostic	20.00	9.05	6.46	100.00 / 100.00
Heart_Beat_ASU (2)	Unconditional	10.00	5.39	3.85	100.00 / 100.00
Status_ASU (18)	Unconditional	30.00	4.66	3.33	100.00 / 100.00
Heart_Beat_Feedback_ASU (17)	Unconditional	10.00	3.94	2.81	100.00 / 100.00
Ins_Disins (1)	Unconditional	20.00	7.59	5.42	100.00 / 100.00

Signals of Unconditional Frame <RLSData>

Signal	Startbit	/	Publisher	Subscribers	Length [Bit]	Init Value	Unit	Encoding
LightSensorFailSts	3	RLS	BCM		1	0		logical
RainSensorFailSts	4	RLS	BCM		1	0		logical
HeadLampActivation	5	RLS	BCM		1	0		logical
WiperMode	6	RLS	BCM		1	0		logical
WiperSpeed	7	RLS	BCM		3	0		logical
LightSensorSelected	10	RLS	BCM		1	0		logical
RainSensorSelected	11	RLS	BCM		1	0		logical
ResponseErrorStsRLS	12	RLS	BCM		1	0		logical
GenericFailStsRLS	13	RLS	BCM		1	0		logical
CurrentFailStsRLS	14	RLS	BCM		1	0		logical
KeepAliveRLS	15	RLS	BCM		1	0		logical

Schedule Table Properties

Name	Operating_Armed
Table Id	0
Bus Load [%]	38.2
Total Cycle Time [ms]	100.0
Resolves collisions f	

Frame Properties

Name	RLSData
Frame ID	41
Size	2
Is Dynamic	False
Publisher	RLS
Subscribers	BCM
CS Model	Enhanced

Click here to view load inconsistencies

LIN Master: BCM Jitter: 0.0 ms Timebase: 10 ms Channel: LIN Speed: 19.200 kbps LIN Protocol: 2.0

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